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Studierichting: Informatica
Oefeningenreeks 1B nr. 42.6

$36z^4 + 13z^2 + 1 = 0$

Stel $t = z^2$

$36t^2 + 13t + 1 = 0$

$$\begin{aligned} D &= 169 - 4 \cdot 36 \cdot 1 \\ &= 169 - 144 \\ &= 25 \end{aligned}$$

$$t_1 = \frac{-13+5}{72} = \frac{-8}{72} = -\frac{i}{9}$$

$$t_2 = \frac{-13-5}{72} = \frac{-18}{72} = -\frac{i}{4}$$

$$\Rightarrow z_1 = \frac{i}{3}$$

$$\Rightarrow z_2 = -\frac{i}{3}$$

$$\Rightarrow z_3 = \frac{i}{2}$$

$$\Rightarrow z_4 = -\frac{i}{2}$$

References